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QUESTION 1:

DRAW A SIMPLE NETWORK DIAGRAM SHOWING THREE DEVICES (LIKE YOUR LAPTOP, A FRIEND'S PHONE, AND A SMART TV) CONNECTED TO A SWITCH, AND LABEL WHICH DEVICE IS IN WHICH VLAN (E.G., VLAN 10 FOR STREAMING, VLAN 20 FOR PERSONAL DEVICES).

ANSWER:

```
          SWITCH
             /
  --------------------------
   /          /          /
LAPTOP      PHONE      SMART TV
VLAN 20     VLAN 20    VLAN 10
```

VLAN DETAILS:

VLAN 10 (STREAMING): SMART TV

VLAN 20 (PERSONAL DEVICES): LAPTOP, PHONE

QUESTION 2:

CONFIGURE A SIMULATED SWITCH USING CISCO PACKET TRACER OR NETSIM: CREATE TWO VLANS (VLAN 10 AND VLAN 20), ASSIGN THREE PCS TO VLAN 10 AND TWO PCS TO VLAN 20, AND VERIFY THAT PCS IN DIFFERENT VLANS CANNOT PING EACH OTHER.

ANSWER:

CREATE VLANS

ENABLE

CONFIGURE TERMINAL

VLAN 10

NAME STREAMING

VLAN 20

NAME PERSONAL

ASSIGN PCS TO VLAN 10

INTERFACE RANGE FA0/1-3

SWITCHPORT MODE ACCESS

SWITCHPORT ACCESS VLAN 10

ASSIGN PCS TO VLAN 20

INTERFACE RANGE FA0/4-5

SWITCHPORT MODE ACCESS

SWITCHPORT ACCESS VLAN 20

VERIFY VLANS

SHOW VLAN BRIEF

PING TEST

PC1 (VLAN 10) → PC2 (VLAN 10) ☑
SUCCESS

PC1 (VLAN 10) → PC4 (VLAN 20) ❌ FAILED

RESULT:

DEVICES IN DIFFERENT VLANS CANNOT COMMUNICATE WITHOUT A ROUTER OR LAYER-3 SWITCH.

QUESTION 3:

IN ANY NETWORK SIMULATOR, SET UP A SCENARIO WHERE YOU MOVE A PC FROM

VLAN 10 TO VLAN 20. SHOW THE BEFORE AND AFTER CONFIGURATION COMMANDS AND TEST CONNECTIVITY TO CONFIRM THE CHANGE.

ANSWER:

BEFORE CHANGE

SHOW VLAN BRIEF

OUTPUT:

FA0/1 VLAN 10

FA0/2 VLAN 10

FA0/3 VLAN 10

MOVE PC ON FA0/3 TO VLAN 20

CONFIGURE TERMINAL

INTERFACE FA0/3

SWITCHPORT MODE ACCESS

SWITCHPORT ACCESS VLAN 20

AFTER CHANGE

SHOW VLAN BRIEF

OUTPUT:

FA0/1 VLAN 10

FA0/2 VLAN 10

FA0/3 VLAN 20

CONNECTIVITY TEST

PC ON FA0/3 CAN NOW COMMUNICATE WITH VLAN 20 DEVICES ☑

PC ON FA0/3 CANNOT COMMUNICATE WITH VLAN 10 DEVICES ✗

RESULT:

THE PC SUCCESSFULLY MOVED FROM VLAN 10 TO VLAN 20.

QUESTION 4:

WRITE A SHORT SCRIPT OR COMMAND LIST THAT WOULD AUTOMATE ASSIGNING 10 PORTS ON A SWITCH TO VLAN 30, USING THE SYNTAX OF YOUR CHOSEN SIMULATOR (PACKET TRACER, NETSIM, OR GNS3).

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ANSWER:

CREATE VLAN 30

VLAN 30

NAME STAFF

ASSIGN 10 PORTS USING INTERFACE RANGE

INTERFACE RANGE FA0/1 - 10

SWITCHPORT MODE ACCESS

SWITCHPORT ACCESS VLAN 30

VERIFY

SHOW VLAN BRIEF

RESULT:

PORTS FA0/1 TO FA0/10 ARE ASSIGNED TO VLAN 30 USING A SINGLE RANGE COMMAND INSTEAD OF CONFIGURING EACH PORT INDIVIDUALLY.